

aday_ogrt_anket_regresyon_modeli_ciktilar

2026-06-15

```
library(readxl)
library(lmtest)
library(sandwich)
library(lmtest)
library(ggplot2)
library(psych)
library(car)
```

```
df <- read_excel("C:/Users/Lenovo/OneDrive/Belgeler/anket_verisi_r_analiz.xlsx")
```

```
df$Cinsiyet <- as.factor(df$Cinsiyet )
df$YerlesimYeri <- as.factor(df$`Buyudugunuz yerlesim yeri`)
df$Brans <- as.factor(df$Brans)
df$Gelir <- as.factor(df$Gelir)
df$Deneyim <- as.factor(df$Deneyim)
df$Sinif <- as.factor(df$Sinif)
df$B1_r <- 6 - df$B1      #ters
df$B5_r <- 6 - df$B5      #ters
df$B6_r <- 6 - df$B6      #ters
df$B7_r <- 6 - df$B7      #ters

b_cols <- paste0("B", 1:7)
c_cols <- paste0("C", 1:9)
d_cols <- paste0("D", 1:6)
f_cols <- paste0("F", 1:10)
e_cols <- paste0("E", 1:9)

df[b_cols] <- lapply(df[b_cols], as.numeric)
df[c_cols] <- lapply(df[c_cols], as.numeric)
df[d_cols] <- lapply(df[d_cols], as.numeric)
df[f_cols] <- lapply(df[f_cols], as.numeric)
df[e_cols] <- lapply(df[e_cols], function(x) ifelse(x == "Evet", 1, 0))
b_cols <- c("B1_r", "B2", "B3", "B4", "B5_r", "B6_r", "B7_r")

df$algi_skor <- rowMeans(df[b_cols])
df$inanc_skor <- rowMeans(df[c_cols])
df$kaygi_skor <- rowMeans(df[d_cols])
df$yeterlik_skor <- rowMeans(df[f_cols])
df$fiziksel_skor <- rowMeans(df[e_cols])
```

```
describe(df$yeterlik_skor)
```

```
##      vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1      1 305 3.56 0.77    3.7    3.63 0.44   1   5     4 -1.15    2.01 0.04
```

```
shapiro.test(df$yeterlik_skor)
```

```
##  
## Shapiro-Wilk normality test  
##  
## data: df$yeterlik_skor  
## W = 0.90254, p-value = 4.03e-13
```

```
model1 <- lm(yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + Brans,  
             data = df)  
summary(model1)
```

```
##  
## Call:  
## lm(formula = yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri +  
##     Brans, data = df)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -2.46226 -0.30924  0.09567  0.44339  1.66116   
##  
## Coefficients:  
##                                     Estimate Std. Error t value  
## (Intercept)                       3.25597     0.24568   13.253  
## CinsiyetKadin                     -0.09608     0.10817    -0.888  
## Sinif2                             0.06143     0.14139     0.434  
## Sinif3                             0.11722     0.14564     0.805  
## Sinif4                             0.19668     0.15560     1.264  
## SinifArtik Yil                     0.34523     0.24572     1.405  
## YerlesimYeriKoy                    -0.12728     0.15748    -0.808  
## YerlesimYeriSehir                  0.02144     0.10107     0.212  
## BransBilgisayar ogretim Teknolojileri ogretmenligi 0.06649     0.26075     0.255  
## BransBiyoloji ogretmenligi         0.36798     0.36705     1.003  
## BransFen Bilgisi ogretmenligi      0.05959     0.29365     0.203  
## BransFizik ogretmenligi            0.49804     0.29171     1.707  
## BransFransiz Dili ogretmenligi     0.47938     0.28811     1.664  
## Bransİlkogretim Matematik ogretmenligi -0.08971     0.27035    -0.332  
## Bransİngiliz Dili ogretmenligi    -0.12122     0.25220    -0.481  
## BransKimya ogretmenligi           0.28094     0.27074     1.038  
## BransMatematik ogretmenligi       0.11385     0.26258     0.434  
## BransOkul oncesi ogretmenligi     0.58604     0.24325     2.409  
## Bransozel Egitim ogretmenligi     0.73129     0.28012     2.611  
## BransRehberlik ve Psikolojik Danismanlik 0.45542     0.24122     1.888  
## BransSinif ogretmenligi           0.25102     0.24441     1.027  
## BransTurkce ogretmenligi          0.14629     0.29259     0.500  
##                                     Pr(>|t|)  
## (Intercept)                       < 2e-16 ***  
## CinsiyetKadin                      0.37517  
## Sinif2                             0.66427  
## Sinif3                             0.42160  
## Sinif4                             0.20726  
## SinifArtik Yil                     0.16112
```

```
## YerlesimYeriKoy 0.41961
## YerlesimYeriSehir 0.83217
## BransBilgisayar ogretim Teknolojileri ogretmenligi 0.79891
## BransBiyoloji ogretmenligi 0.31695
## BransFen Bilgisi ogretmenligi 0.83933
## BransFizik ogretmenligi 0.08886 .
## BransFransiz Dili ogretmenligi 0.09725 .
## Bransİlkogretim Matematik ogretmenligi 0.74028
## Bransİngiliz Dili ogretmenligi 0.63114
## BransKimya ogretmenligi 0.30031
## BransMatematik ogretmenligi 0.66491
## BransOkul oncesi ogretmenligi 0.01662 *
## Bransozel Egitim ogretmenligi 0.00952 **
## BransRehberlik ve Psikolojik Danismanlik 0.06006 .
## BransSinif ogretmenligi 0.30528
## BransTurkce ogretmenligi 0.61747
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7482 on 283 degrees of freedom
## Multiple R-squared:  0.1126, Adjusted R-squared:  0.0468
## F-statistic: 1.711 on 21 and 283 DF,  p-value: 0.02868
```

```
shapiro.test(residuals(model1))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(model1)
## W = 0.92764, p-value = 5.223e-11
```

```
bptest(model1)
```

```
##
## studentized Breusch-Pagan test
##
## data: model1
## BP = 28.106, df = 21, p-value = 0.1372
```

```
dwtest(model1)
```

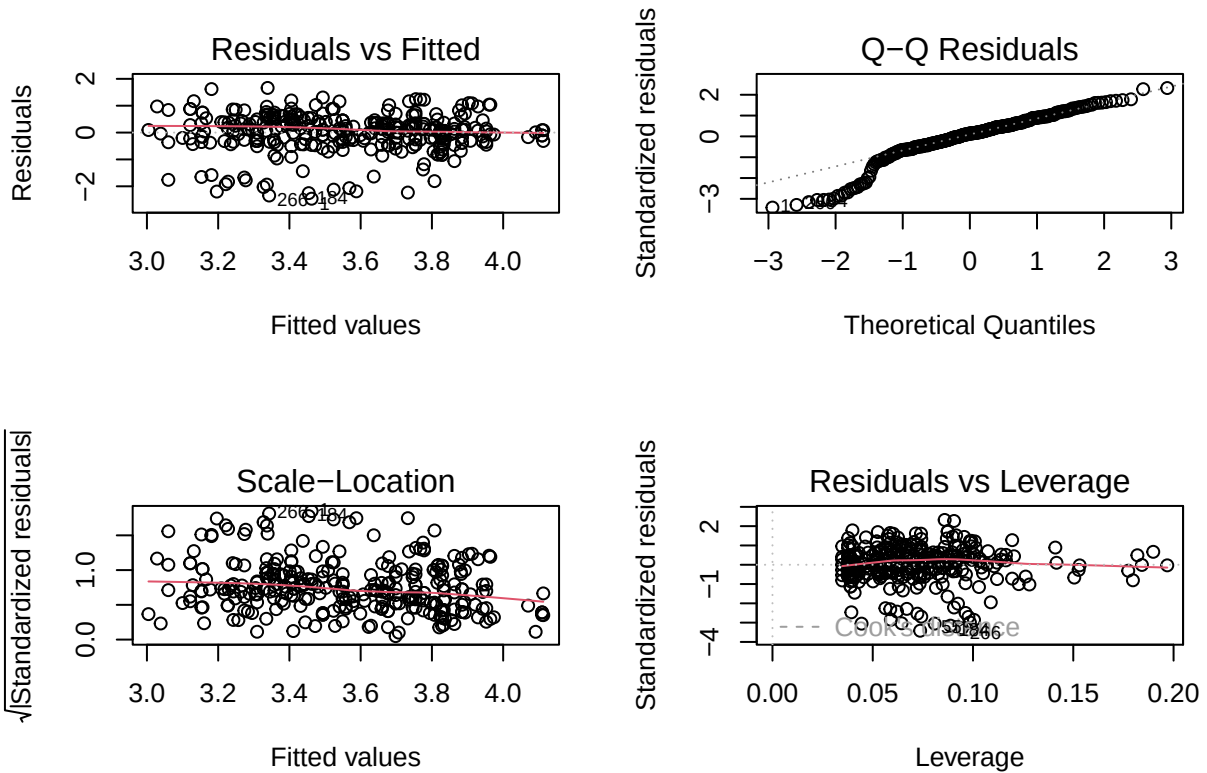
```
##
## Durbin-Watson test
##
## data: model1
## DW = 1.9622, p-value = 0.3266
## alternative hypothesis: true autocorrelation is greater than 0
```

```
vif(model1)
```

```
## GVIF Df GVIF^(1/(2*Df))
```

```
## Cinsiyet      1.182216  1      1.087298
## Sinif         1.409532  4      1.043841
## YerlesimYeri 1.172361  2      1.040556
## Brans         1.786853 14      1.020947
```

```
par(mfrow = c(2, 2))
plot(model1)
```



```
model2 <- lm(yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + Brans +
             algi_skor + inanc_skor + kaygi_skor + fiziksel_skor,
             data = df)
summary(model2)
```

```
##
## Call:
## lm(formula = yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri +
##     Brans + algi_skor + inanc_skor + kaygi_skor + fiziksel_skor,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.65630 -0.33926 -0.00565  0.38063  1.32254
##
## Coefficients:
##                                     Estimate Std. Error t value
```

## (Intercept)	1.15992	0.35565	3.261
## CinsiyetKadin	-0.18290	0.07826	-2.337
## Sinif2	0.14331	0.10087	1.421
## Sinif3	0.01779	0.10418	0.171
## Sinif4	0.13361	0.11207	1.192
## SinifArtik Yil	0.28309	0.17709	1.599
## YerlesimYeriKoy	-0.16630	0.11449	-1.452
## YerlesimYeriSehir	0.07230	0.07197	1.005
## BransBilgisayar ogretim Teknolojileri ogretmenligi	-0.21854	0.18757	-1.165
## BransBiyoloji ogretmenligi	0.15299	0.26389	0.580
## BransFen Bilgisi ogretmenligi	-0.05724	0.21017	-0.272
## BransFizik ogretmenligi	0.13043	0.21343	0.611
## BransFransiz Dili ogretmenligi	0.27316	0.20702	1.320
## Bransİlkogretim Matematik ogretmenligi	-0.33155	0.19364	-1.712
## Bransİngiliz Dili ogretmenligi	-0.08992	0.18144	-0.496
## BransKimya ogretmenligi	-0.09810	0.19374	-0.506
## BransMatematik ogretmenligi	-0.05230	0.18817	-0.278
## BransOkul oncesi ogretmenligi	0.14436	0.17789	0.812
## Bransozel Egitim ogretmenligi	0.21078	0.20687	1.019
## BransRehberlik ve Psikolojik Danismanlik	0.05914	0.17565	0.337
## BransSinif ogretmenligi	-0.01688	0.17511	-0.096
## BransTurkce ogretmenligi	-0.11025	0.21065	-0.523
## algi_skor	-0.15838	0.07997	-1.981
## inanc_skor	0.81235	0.05062	16.047
## kaygi_skor	-0.04252	0.03978	-1.069
## fiziksel_skor	0.10626	0.26866	0.396
##	Pr(> t)		
## (Intercept)	0.00125	**	
## CinsiyetKadin	0.02014	*	
## Sinif2	0.15650		
## Sinif3	0.86454		
## Sinif4	0.23417		
## SinifArtik Yil	0.11106		
## YerlesimYeriKoy	0.14749		
## YerlesimYeriSehir	0.31599		
## BransBilgisayar ogretim Teknolojileri ogretmenligi	0.24497		
## BransBiyoloji ogretmenligi	0.56256		
## BransFen Bilgisi ogretmenligi	0.78556		
## BransFizik ogretmenligi	0.54164		
## BransFransiz Dili ogretmenligi	0.18808		
## Bransİlkogretim Matematik ogretmenligi	0.08797	.	
## Bransİngiliz Dili ogretmenligi	0.62056		
## BransKimya ogretmenligi	0.61300		
## BransMatematik ogretmenligi	0.78127		
## BransOkul oncesi ogretmenligi	0.41775		
## Bransozel Egitim ogretmenligi	0.30912		
## BransRehberlik ve Psikolojik Danismanlik	0.73660		
## BransSinif ogretmenligi	0.92329		
## BransTurkce ogretmenligi	0.60113		
## algi_skor	0.04863	*	
## inanc_skor	< 2e-16	***	
## kaygi_skor	0.28605		
## fiziksel_skor	0.69277		
## ---			

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.531 on 279 degrees of freedom
## Multiple R-squared:  0.5594, Adjusted R-squared:  0.5199
## F-statistic: 14.17 on 25 and 279 DF,  p-value: < 2.2e-16
```

```
anova(model1, model2)
```

```
## Analysis of Variance Table
##
## Model 1: yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + Brans
## Model 2: yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + Brans + algi_skor +
##          inanc_skor + kaygi_skor + fiziksel_skor
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1      283 158.425
## 2      279  78.661   4    79.764 70.728 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
shapiro.test(residuals(model2))
```

```
##
## Shapiro-Wilk normality test
##
## data:  residuals(model2)
## W = 0.99478, p-value = 0.3883
```

```
bptest(model2)
```

```
##
## studentized Breusch-Pagan test
##
## data:  model2
## BP = 36.516, df = 25, p-value = 0.06413
```

```
dwtest(model2)
```

```
##
## Durbin-Watson test
##
## data:  model2
## DW = 1.8583, p-value = 0.08594
## alternative hypothesis: true autocorrelation is greater than 0
```

```
vif(model2)
```

```
##              GVIF Df GVIF^(1/(2*Df))
## Cinsiyet      1.228515  1      1.108384
## Sinif         1.546349  4      1.055999
## YerlesimYeri  1.237184  2      1.054651
```

```
## Brans          2.454245 14          1.032585
## algi_skor      1.589686  1          1.260828
## inanc_skor     1.369633  1          1.170313
## kaygi_skor     1.485971  1          1.219004
## fiziksel_skor  1.185288  1          1.088709
```

```
model_bos <- lm(yeterlik_skor ~ 1, data = df)

model_final <- step(model2,
  direction = "both",
  trace = 1)
```

```
## Start: AIC=-361.33
## yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + Brans + algi_skor +
##      inanc_skor + kaygi_skor + fiziksel_skor
##
##              Df Sum of Sq    RSS    AIC
## - Brans       14      5.931  84.592 -367.16
## - Sinif        4      1.476  80.137 -363.65
## - fiziksel_skor 1      0.044  78.705 -363.15
## - kaygi_skor   1      0.322  78.983 -362.08
## <none>                78.661 -361.33
## - YerlesimYeri  2      1.508  80.169 -359.53
## - algi_skor     1      1.106  79.767 -359.07
## - Cinsiyet      1      1.540  80.201 -357.41
## - inanc_skor    1     72.599 151.260 -163.90
##
## Step: AIC=-367.16
## yeterlik_skor ~ Cinsiyet + Sinif + YerlesimYeri + algi_skor +
##      inanc_skor + kaygi_skor + fiziksel_skor
##
##              Df Sum of Sq    RSS    AIC
## - Sinif        4      1.240  85.832 -370.72
## - fiziksel_skor 1      0.105  84.697 -368.78
## - kaygi_skor    1      0.134  84.725 -368.67
## - algi_skor     1      0.391  84.982 -367.75
## - YerlesimYeri  2      1.020  85.611 -367.50
## <none>                84.592 -367.16
## - Cinsiyet      1      0.994  85.586 -365.59
## + Brans        14      5.931  78.661 -361.33
## - inanc_skor    1     78.402 162.993 -169.11
##
## Step: AIC=-370.72
## yeterlik_skor ~ Cinsiyet + YerlesimYeri + algi_skor + inanc_skor +
##      kaygi_skor + fiziksel_skor
##
##              Df Sum of Sq    RSS    AIC
## - kaygi_skor    1      0.162  85.994 -372.14
## - fiziksel_skor 1      0.196  86.028 -372.02
## - YerlesimYeri  2      0.780  86.612 -371.96
## - algi_skor     1      0.340  86.172 -371.51
## <none>                85.832 -370.72
## - Cinsiyet      1      1.184  87.016 -368.54
## + Sinif         4      1.240  84.592 -367.16
```

```

## + Brans          14      5.695  80.137 -363.65
## - inanc_skor      1     79.540 165.372 -172.70
##
## Step:  AIC=-372.14
## yeterlik_skor ~ Cinsiyet + YerlesimYeri + algi_skor + inanc_skor +
##      fiziksel_skor
##
##           Df Sum of Sq      RSS      AIC
## - fiziksel_skor  1      0.145  86.139 -373.63
## - algi_skor      1      0.213  86.207 -373.39
## - YerlesimYeri   2      0.822  86.816 -373.24
## <none>                                85.994 -372.14
## + kaygi_skor     1      0.162  85.832 -370.72
## - Cinsiyet       1      1.392  87.386 -369.24
## + Sinif          4      1.268  84.725 -368.67
## + Brans          14      5.499  80.494 -364.30
## - inanc_skor      1     84.280 170.273 -165.79
##
## Step:  AIC=-373.63
## yeterlik_skor ~ Cinsiyet + YerlesimYeri + algi_skor + inanc_skor
##
##           Df Sum of Sq      RSS      AIC
## - algi_skor      1      0.305  86.443 -374.55
## - YerlesimYeri   2      0.877  87.016 -374.54
## <none>                                86.139 -373.63
## + fiziksel_skor  1      0.145  85.994 -372.14
## + kaygi_skor     1      0.111  86.028 -372.02
## - Cinsiyet       1      1.368  87.507 -370.82
## + Sinif          4      1.345  84.794 -370.43
## + Brans          14      5.573  80.566 -366.03
## - inanc_skor      1     86.360 172.498 -163.83
##
## Step:  AIC=-374.55
## yeterlik_skor ~ Cinsiyet + YerlesimYeri + inanc_skor
##
##           Df Sum of Sq      RSS      AIC
## - YerlesimYeri   2      0.811  87.254 -375.70
## <none>                                86.443 -374.55
## + algi_skor      1      0.305  86.139 -373.63
## + fiziksel_skor  1      0.237  86.207 -373.39
## + kaygi_skor     1      0.005  86.439 -372.57
## - Cinsiyet       1      1.496  87.939 -371.32
## + Sinif          4      1.300  85.143 -371.17
## + Brans          14      5.064  81.379 -364.96
## - inanc_skor      1     91.927 178.370 -155.62
##
## Step:  AIC=-375.7
## yeterlik_skor ~ Cinsiyet + inanc_skor
##
##           Df Sum of Sq      RSS      AIC
## <none>                                87.254 -375.70
## + fiziksel_skor  1      0.286  86.968 -374.70
## + YerlesimYeri   2      0.811  86.443 -374.55
## + algi_skor      1      0.239  87.016 -374.54

```

```
## + kaygi_skor      1      0.016  87.238 -373.76
## - Cinsiyet        1      1.433  88.687 -372.74
## + Sinif           4      1.053  86.201 -371.41
## + Brans           14     4.731  82.524 -364.70
## - inanc_skor      1     91.233 178.488 -159.42
```

```
shapiro.test(residuals(model_final))
```

```
##
## Shapiro-Wilk normality test
##
## data: residuals(model_final)
## W = 0.99584, p-value = 0.5991
```

```
bptest(model_final)
```

```
##
## studentized Breusch-Pagan test
##
## data: model_final
## BP = 7.7503, df = 2, p-value = 0.02075
```

```
dwtest(model_final)
```

```
##
## Durbin-Watson test
##
## data: model_final
## DW = 1.8695, p-value = 0.1255
## alternative hypothesis: true autocorrelation is greater than 0
```

```
vif(model_final)
```

```
## Cinsiyet inanc_skor
## 1.022409 1.022409
```

```
coeftest(model_final, vcov = vcovHC(model_final, type = "HC3"))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  0.778276   0.221308  3.5167  0.000504 ***
## CinsiyetKadin -0.160937   0.075080 -2.1435  0.032869 *
## inanc_skor    0.786803   0.056262 13.9847 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(model_final)
```

```
##
## Call:
## lm(formula = yeterlik_skor ~ Cinsiyet + inanc_skor, data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.58971 -0.36579  0.01029  0.33678  1.59905
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.77828    0.16721   4.654 4.87e-06 ***
## CinsiyetKadin -0.16094    0.07227  -2.227  0.0267 *
## inanc_skor     0.78680    0.04428  17.770 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5375 on 302 degrees of freedom
## Multiple R-squared:  0.5113, Adjusted R-squared:  0.508
## F-statistic: 158 on 2 and 302 DF, p-value: < 2.2e-16
```